12TH CHEMISTRY · CHAPTER 28 · CHAPTER TEST

Chromatography

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 28	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. The stationary phase in paper chromatography is:

(a) The solvent (b) Helium
(c) Silica (d) Water in paper

2. The Rf value has:

(a) Unit cm (b) No unit
(c) Unit min (d) Unit g

3. The locating agent for amino acids is:

(a) Ninhydrin (b) Benzene
(c) Iodine (d) Alumina

4. The mobile phase in GC is:

(a) Water (b) Ethanol
(c) An inert gas (d) Silica

5. The time taken to pass through a GC column is:

(a) Rf value (b) Retention time
(c) Baseline (d) Peak area

6. In TLC, polar substances:

(a) Move fastest (b) Evaporate
(c) Move slowly (d) Do not adsorb

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. What is adsorption? Give one example.
ii. Define Rf value. Why has it no unit?
iii. Why is the baseline drawn with pencil?
iv. State two limitations of TLC.
v. What is the function of the carrier gas in GC?
vi. State two limitations of gas chromatography.

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Describe the principle and working of thin layer chromatography.
(b) Describe the working of gas chromatography.

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